END-TO-END PROBABILISTIC FLOOD FORECASTING & EXPOSURE PIPELINE

Operational 4-Stage Architecture: Embargoed Meteorological Ingestion → County ML → High-Resolution Pixel Downscaling → Asset Overlay

STAGE 1: DATA INGESTION

Strict Friday Embargo (t - 3 Days)

- **Meteorology & Hydrology**
  - ERA5 Daily Precipitation (tp)
  - ERA5 Surface Runoff (ro)
  - GLEAM Evapotranspiration (ET)
  - Soil Moisture & 7d/14d Deficit
- **Satellite Altimetry & Gauges**
  - Hydroweb: Lake Victoria Levels
  - DAHITI: Lake Albert / Kyoga Altimetry
  - Dartmouth (DFO) River Discharge
- **Ground Truth Flood Target**
  - NASA MODIS 8-day Flood Masks
  - 26-Year Climatology (2000-2025)
  - Recurring & Unusual Flood Masks

AUDIT: EMBARGO
Strict Friday 00:00 UTC cutoff.
Zero lookahead leakage across weeks.

STAGE 2: TASK A (LIGHTGBM)

County Probabilistic ML Engine

- **Head 1: Detection Probability**
  - Binary Logloss Classifier (det_prob)
  - Brier Score: 0.0650 (Calibrated)
  - +34.8% Skill vs Climatology
  - +21.7% Skill vs Persistence
- **Head 2: Quantile Area (km²)**
  - Asymmetric Pinball Quantile Loss
  - q10, q50 (Median), q90 (Worst)
  - Flood-Week MAE: 8.68 km²
- **Head 3: Multi-Week Duration**
  - Persistence 1 to 4 weeks ahead
  - Independent multi-horizon heads

AUDIT: VALIDATION
Walk-forward temporal splits.
No test season ever seen in training.

STAGE 3: TASK B (DOWNSCALER)

Physical Spatial Allocation (~232m)

- **Spatial Signature Prior**
  - Empirical freq_given(p, c, m)
  - 584,313 canonical cells
  - Native ~232 m MODIS grid
- **Coordinate Snapping Bugfix**
  - Index: k = round(v/step - 0.5)
  - 100% cell merge (Grid vs Parquet)
  - Zero floating cell artifacts
- **Conservation Constraints**
  - Exact Area Conservation Matching q50
  - Probability Map: det_prob × prior
  - Severity Layer: scaled to q50

AUDIT: CONSERVATION
Nov 4, 2024 peak flood:
q50 = 166.32 km² == Implied map area.

STAGE 4: ASSET EXPOSURE

Humanitarian Impact & Early Action

- **Staple Crop Phenology**
  - ASAP Sorghum & Pearl Millet Masks
  - Growth stages vs Flood timeline
  - Critical Harvest Window (Sep-Nov)
- **Livestock & Pastoral Rangeland**
  - FAO Cattle Density Grids
  - Pastoral transhumance routes
  - Rangeland submergence alerts
- **Population & Health Facilities**
  - WorldPop 100m Population Density
  - Health facility accessibility
  - Pre-emptive action bulletins

AUDIT: HUMANITARIAN
Actionable 1-4 week early warnings
to protect ripening grain & cattle.